Apache JMeter

Introduction

- What is Jmeter?
- Install and Start
- Understand GUI
- Test Plan
- Thread Group
- Samplers
- Listeners
- Recording
- CLI
- Best Practices (Before-During- After)

What is JMeter?

- Performance test application
- Build using Java (SDK required)
- Free & Open Source
- Recording
- CLI (command Line Interface)
- Reports

How to Install JMeter?

Windows | Mac | Linux

- **Step 1:** Check Java is installed in your system

  java -version

- **Step 2:** Download JMeter from https://jmeter.apache.org/
- **Step 3:** unzip and keep JMeter folder any location
- **Step 4:** start JMeter

  Windows: Jmeter/bin- jmeter.bat

  Mac: open Terminal jmeter/bin – jmeter.sh

JMeter Overview:

File -> Template -> Building a web test plan

Understand GUI

- **Menu bar**: Access various menus like File, Edit, View and Help
- **Toolbar**: Quick access to commonly used actions such as New, Open, Save and Run
- **Test Plan**: The main area where you'll create and configure your test plan
  - **Tree View**: A hierarchical view of your test plan, showing the structure of your test
  - **Results Tree**: Display the results of your test execution
- **Editor**: Select an element from test plan and work on it in Editor

Time, Logs, Users

How to Create JMeter Test Plan?

A test plan describes a series of

steps JMeter will execute when run.

- ✓ Test your website or application
- ✓ How many users to simulate
- ✓ What actions they should perform

https://cse.buet.ac.bd/

Protocol: https

Server Name: cse.buet.ac.bd

Path: /

Steps:

- **Step 1:** Start JMeter
- **Step 2:** Create a Test Plan
- **Step 3:** Create a Thread Group (Users)
- **Step 4:** Add a Sampler (HTTP Request)
- **Step 5:** Add Listeners

  (add View result in Tree,

  add View Result in Table)
- **Step 6:** Run the Test

Ramp-up periods: How much time it will take to create all the users? Uniformly distributed

Dissection of a Web Address

https://www.google.com/search?q=blazemeter+chrome+extension&sca_esv=314db17b76bb14ae&sxsrf=ANbLn45sKUgiBTDwijbMLikiWF2jB_LCg%3A1774693282708&source=hp&ei=oqvHadOIKbuXwcsP9rXzqAk&iflsig=AFdpzrgAAAAAace5snqZTNEy1vfKv35Kpg8JAz7TlAvO&oq=&sclient=gws-wiz

Dissection:

- **HTTP Protocol:** https
- **Server Name:** www.google.com
- **Path:** /search
- **Parameter Variables and Values:**
  - q: blazemeter+chrome+extension
  - sca_esv: 314db17b76bb14ae
  - sxsrf: ANbL-n45sKUgiBTDwijbMLikiWF2jB_LCg%3A1774693282708
  - source: hp
  - ei: oqvHadOIKbuXwcsP9rXzqAk
  - iflsig: AFdpzrgAAAAAace5snqZTNEy1vfKv35Kpg8JAz7TlAvO
  - oq: (empty)
  - sclient: gws-wiz

Thread Group

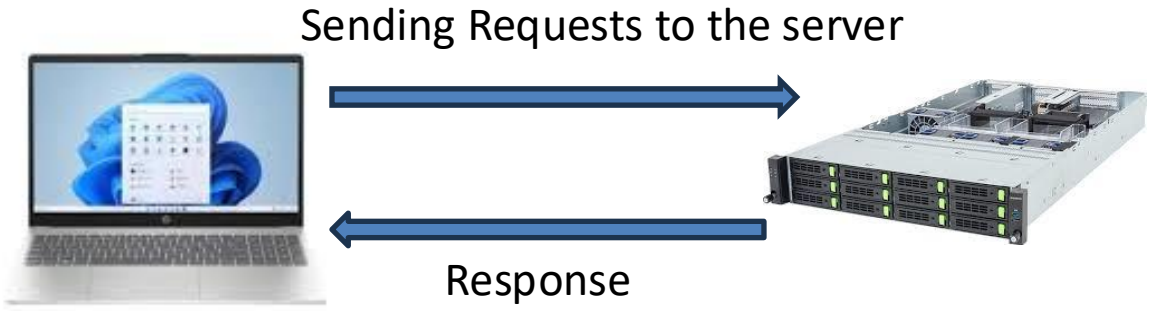
Simulating multiple users: how many virtual users will access your website or application during the test.

Step 1: Right click on the test plan,

go to Add -> Threads (Users) -> Thread group

Step 2: Check the fields and configuration of Thread group

Samplers



Step 1: Right click on Thread Group,

go to Add -> Sampler

Step 2: Add HTTP Request Sampler

Step 3: Configure a Sampler to send a request to a website (e.g., Google)

Step 4: Check other Samplers

Some Sampler Names:

- ❑ HTTP Request
- ❑ Debug Sampler
- ❑ BeanShell Sampler
- ❑ FTP Request
- ❑ JDBC Request
- ❑ JUnit Request
- ❑ Java Request
- ❑ LDAP Request
- ❑ Mail Reader Sampler

Listeners (Reporting)

Used for Reporting (Alpha Test)

Listener: Capturing and viewing test results

Latency = time to get the first byte of response from server to client (starting to response)

Sample Time: Response time (Getting complete response)

0 ms ----------1000 ms ---------- 1200ms ----------2000ms

Send Request Connect Time Latency Time Sample Time

Step 1: Right click on Thread Group or Sampler,

go to Add -> Listener-> View Results Tree

Step 2: Can add more listeners

Step 3: Run the test and check results

Some Listeners:

- View Results in Table
- View Results Tree
- Aggregate Report
- Graph Results
- Summary Report
- Simple Data Writer

Assertions

Assertions: Checks on the request / response

- **Response Assertion**
- **Duration Assertion**
- **Size Assertion**
- **HTML Assertion**
- **JSON Assertion**
- **XPATH Assertion**

Test Script Recording

Recorder is a **Proxy** Server that will capture the traffic

Step 1: Add Recording Controller(Add Logic Controller -> Recording Controller)

Recorded Samplers are placed in Recording Controllers (add More Recording Controller & Rename)

Two types of Recording: Manual & Template

Step 2: Add Test Script Recorder

Test Plan -> Add -> Non-Test Elements -> HTTP(s) Test Script Recorder

Configure the Recorder

Port: Choose a port for the proxy server (e.g., 8888)

Target Controller: HTTP(s) -> Test Plan Creation -> Target Controller

Requests Filtering: Exclude any unwanted requests (e.g., image, CSS)

Step 3: Set Brower to use JMeter as a proxy on the specific port.

Chrome: Three dots -> Settings (Search Proxy) -> Open your computer's proxy setting -> set manual proxy (localhost, 8888)

Step 4: Set Brower to use JMeter as a proxy on the specific port.

Chrome: Three dots -> settings (search certificate) -> security -> Manage certificate -> add JMeter Certificate (Trusted Publishers)

Click **start** to generate JMeter Certificate then **Stop**

More on Test Script Recording

- ✓ Chrome is more secured browser than Firefox. So, if you face problem to work with Chrome, you can use **Firefox**.
- ✓ After setting manual proxy on browser, **Internet** will not work without starting JMeter Recorder. So, **Add certificate** before changing proxy.
- ✓ **Recorded data** is found in respective **Recording Controller**. To Record data, start JMeter recorder by clicking **start** and then browse the respective website.
- ✓ **Recording login Demo website**: Search "Orangehrm demo" or use link https://opensource-demo.orangehrmlive.com/web/index.php/auth/login Learn how to use template for Test Recording.
- ✓ After Recording, rename actions, add Listeners and run JMeter Test on the recorded data.

Command Line Interface (CLI)

GUI Consumes Memory, Slower

- CLI integrates with external process CI CD
  - How to run JMeter from CLI
  - How to log Results
  - How to see command line helps and options
  - How to run from any location on your system (add in path env variables)

Step 1: Open command line-> Go to JMeter's bin folder -> Run commands

jmeter –n –t "location of test plan file" –l "location of result file" –e –o "location of report folder"

Symbol	Meaning
n	No GUI mode
t	Test plan
l	Log Results
e	Generate Reports (HTML Report)
o	Output Folder (where the report will be stored)
g	Generate a Report Dashboard only (HTML Report)
h	Help (Options for commands)

Result file extension may be .jtl (opened in jmeter listener), .csv, etc.

Command Line Interface (CLI)

Example Commands:

- jmeter –n –t “C:\Users\Administrator\Desktop\Tests\Testq.jmx” –l “C:\Users\Administrator\Desktop\Tests\Reports\Results.jtl”

- jmeter –n –t “C:\Users\Administrator\Desktop\Tests\Testq.jmx” –l “C:\Users\Administrator\Desktop\Tests\Reports\Results.jtl” –e –o “C:\Users\Administrator\Desktop\Tests\Reports\HTML Report”

- jmeter –g “C:\Users\Administrator\Desktop\Tests\Reports\Results.jtl” –o “C:\Users\Administrator\Desktop\Tests\Reports\HTML Report”

Complete Task to Generate HTML Report

- ✓ Use of GUI to Generate Test Plan

  Test Plan -> Thread Group -> Sampler (HTTP Request) -> set protocol, server name, path (/moodle/, /home/, /notice/) -> Rename HTTP Request

  Test plan -> Config Element -> HTTP Cookie Manger, HTTP Cache Manager

  Test plan -> Listener -> View Result Tree => Run

  Test plan -> Assertions -> Duration Assertion => Save

- ✓ Use of CLI to Generate Report

  jmeter –n –t "location of test plan file" –l "location of result file" –e –o "location of report folder"

  or

  jmeter –n –t "location of test plan file" –l "location of result file"

  jmeter –g "location of result file" –o "location of report folder"

Some Real World Examples

Scenario 1: **Basic Load Test from CLI**
Simulate 50 Users concurrently accessing OrnageHRM Login page for 5 minutes.

jmeter –n –t OrangeHRM.jmx –l loginResults.jtl –Jthreads = 50 – Jduration=300

Scenario 2: **Running a specific Test Fragment**
Execute only a particular part of Test Plan (e.g., login process)

jmeter –n –t OrangeHRM_Full.jmx –l loginResults.jtl –T Login_Thread_Group

Scenario 3: **Distributed Load Test**
Generate a higher load on the website server by running test from multiple machines (distributed testing)

jmeter –n –t OrangeHRM.jmx –r –R server1, server2, –l masterResults.jtl

Best Practices

Before:

- ✓ Get approvals and authorizations from all stakeholders.
- ✓ Understand the Application
- ✓ Requirements for the Test- What to test, scenarios
- ✓ Understand the configuration of client machine requirements
- ✓ Environment: Test Environment vs. Production Environment. Always run with the same infrastructure, network and resources for consistent results and comparing with baseline
- ✓ Test Data – how to clean them after the Test
- ✓ Expected behavior or performance or baseline. Always have a baseline for comparison.
- ✓ Reporting format

Best Practices

During:

- ✓ Do not use Test Environment for other activities.
- ✓ Create more realistic Test. Add Think time.
- ✓ Add some ramp-up time … Give some breathing time.
- ✓ Monitor the progress
- ✓ Monitor the Client and Server Resources (CPU, Memory, Network, etc)
- ✓ Communicate with team if you see any issue

After:

- ✓ Analyze and document the Results.
- ✓ Create Reporting Formats to share with team and stakeholders
- ✓ Collaborate with team
- ✓ Document your learning for yourself and others